MSG No. 94.134 $P_I4_22_12$ [Type IV, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4'2'2

No.	position	mapping
1	$[0, 0, 0]$	[1,6,7,8,10,11,12,13]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,4,5,9,14,15,16]

Table 2: Wyckoff site: 2b, site symmetry: 4'2'2

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,6,7,8,10,11,12,13]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,3,4,5,9,14,15,16]

Table 3: Wyckoff site: 4c, site symmetry: 22'2'.

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,6,12,13]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,15,16]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[4,5,9,14]
4	$\left[\frac{1}{2}, 0, 0\right]$	[7,8,10,11]

Table 4: Wyckoff site: 4d, site symmetry: 2.2'2'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[1,6,15,16]
2	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[2,3,12,13]
3	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[4,5,10,11]
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[7,8,9,14]

Table 5: Wyckoff site: 4e, site symmetry: 4'..

No.	position	mapping
1	$[0, 0, z]$	[1,6,10,11]
2	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[2,3,9,14]
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[4,5,15,16]
4	$[0, 0, -z]$	[7,8,12,13]

Table 6: Wyckoff site: 8f, site symmetry: 2..

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,6]
2	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[2,3]
3	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[4,5]

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,8]`
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[\frac{1}{2},\ 0,\ z\right]$	`[10,11]`
7	$\left[0,\ \frac{1}{2},\ -z\right]$	`[12,13]`
8	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[15,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[2,5]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,4]`
4	$[-x,\ -x,\ 0]$	`[6,8]`
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,15]`
6	$[-x,\ x,\ 0]$	`[10,13]`
7	$[x,\ -x,\ 0]$	`[11,12]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `.2'.`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,12]`
2	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[2,15]`
3	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,16]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,9]`
5	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,14]`
6	$[-x,\ 0,\ 0]$	`[6,13]`
7	$[0,\ x,\ 0]$	`[7,10]`
8	$[0,\ -x,\ 0]$	`[8,11]`

Table 9: Wyckoff site: `8i`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,12]`
2	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[2,15]`
3	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[3,16]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,9]`
5	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[5,14]`
6	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[6,13]`
7	$\left[0,\ x,\ \frac{1}{2}\right]$	`[7,10]`
8	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[8,11]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	`[1,15]`
2	$\left[-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	`[2,13]`
3	$\left[x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	`[3,12]`
4	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{4}\right]$	`[4,11]`
5	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{4}\right]$	`[5,10]`
6	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	`[6,16]`
7	$\left[x+\frac{1}{2},\, x,\, \frac{3}{4}\right]$	`[7,9]`
8	$\left[\frac{1}{2}-x,\, -x,\, \frac{3}{4}\right]$	`[8,14]`

Table 11: Wyckoff site: `16k`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[2]`
3	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[3]`
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	`[4]`
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[5]`
6	$[-x,\, -y,\, z]$	`[6]`
7	$[y,\, x,\, -z]$	`[7]`
8	$[-y,\, -x,\, -z]$	`[8]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[9]`
10	$[-y,\, x,\, z]$	`[10]`
11	$[y,\, -x,\, z]$	`[11]`
12	$[x,\, -y,\, -z]$	`[12]`
13	$[-x,\, y,\, -z]$	`[13]`
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[14]`
15	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[15]`
16	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	`[16]`